PROJECT PRE-FINAL CHECK-IN

Mitigating Dialect Bias in Toxicity Detection

Formal Problem Definition: How can we build a toxicity detector (binary classifier) that accurately identifies toxic language without unfairly flagging African American English (AAE) as toxic/offensive?

(Classes: Toxic vs Non-Toxic, Groups: AAE vs SAE)

Goal: Reduce Dialect Biasness while maintaining overall model performance for toxicity detection. (Analyze tradeoffs: Fairness vs Accuracy)

Dialect Bias in Dataset

category	count	AAE corr.
hate speech	1,430	−0.057
offensive	19,190	0.420
none	4,163	−0.414
total	**24,783**	

Davidson et al., 2017 TwitterAAE dataset includes annotations of 25K tweets as hate speech, offensive, or none. The authors collected data from Twitter crowdsourced at least three annotations per tweet.

The Risk of Racial Bias in Hate Speech Detection

Models learn spurious correlations between AAE language patterns and toxicity due to data bias

Very Small Negative Correlation

Strong Positive Correlation

- AAE text is strongly associated with "offensive" label

Strong Negative Correlation

- AAE text is less likely to be labeled as "Non-Toxic"

Interpretation: AAE text is more likely to be labeled as offensive even when it is not toxic

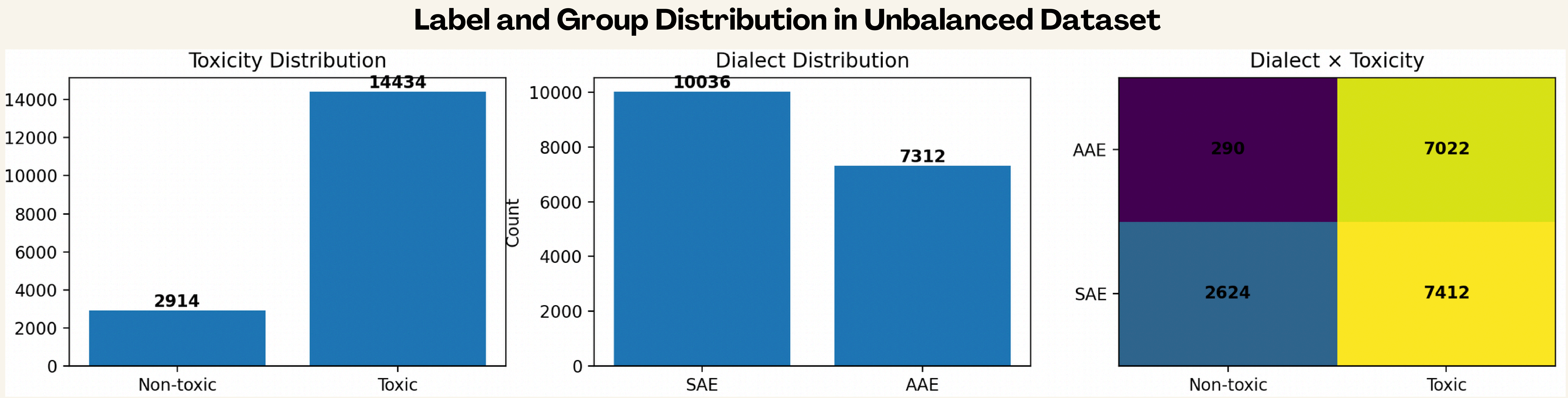
Dataset Preprocessing and Exploration

Davidson Hate Speech Dataset : Contains labeled tweets categorized as Hate speech, Offensive language, or Neither

Used **TwitterAAE** (**Blodgett et al.**) **:** Assigns probability of AAE dialect. The model is trained on more than 12 million tweets.

Used two different thresholds for predicting dialect on tweets. **Strict**: p_AAE ≥ 0.8, **Relaxed**: p_AAE ≥ 0.6

Unbalanced Dataset

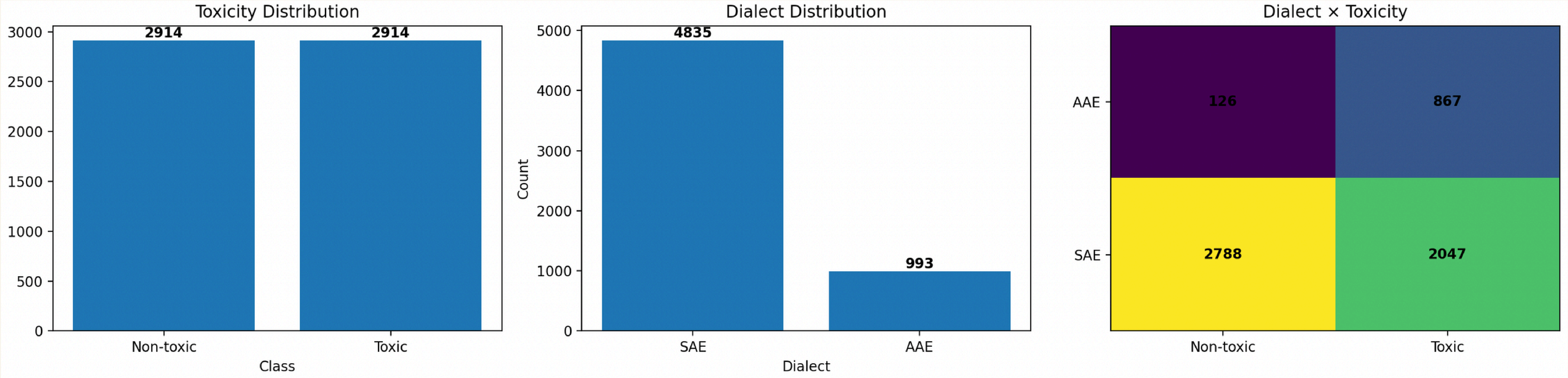


- **Problem:**
  - Original dataset has label imbalance (toxic vs non-toxic)
  - AAE is strongly correlated with toxic labels
  - Label imbalance can mask true source of bias

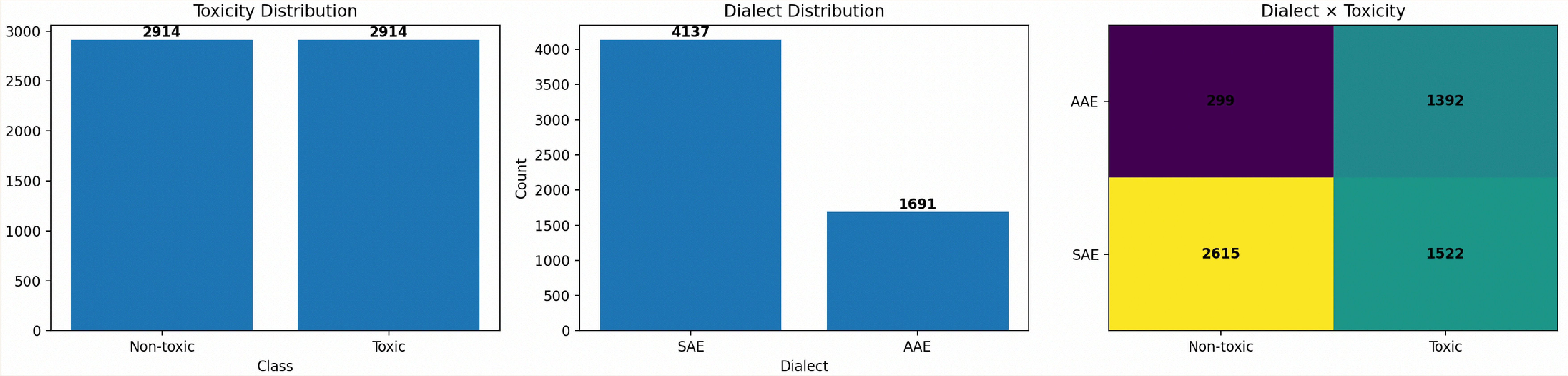
- **Solution:**
  - Balance toxic and non-toxic samples
  - Ensures fairness evaluation reflects dialect bias, not class imbalance

Balanced Dataset

Strict Threshold = 0.8

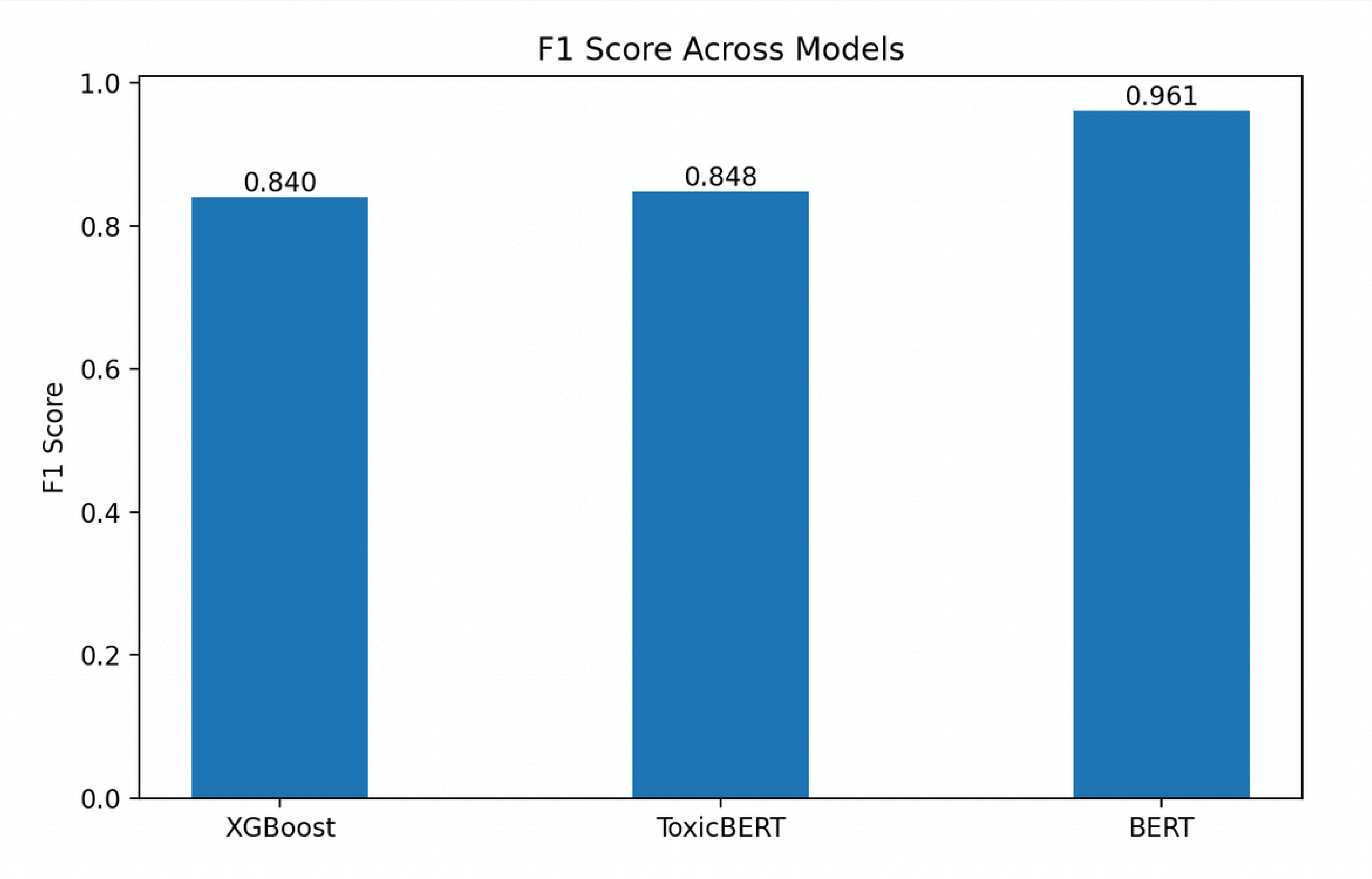


Relaxed Threshold = 0.6

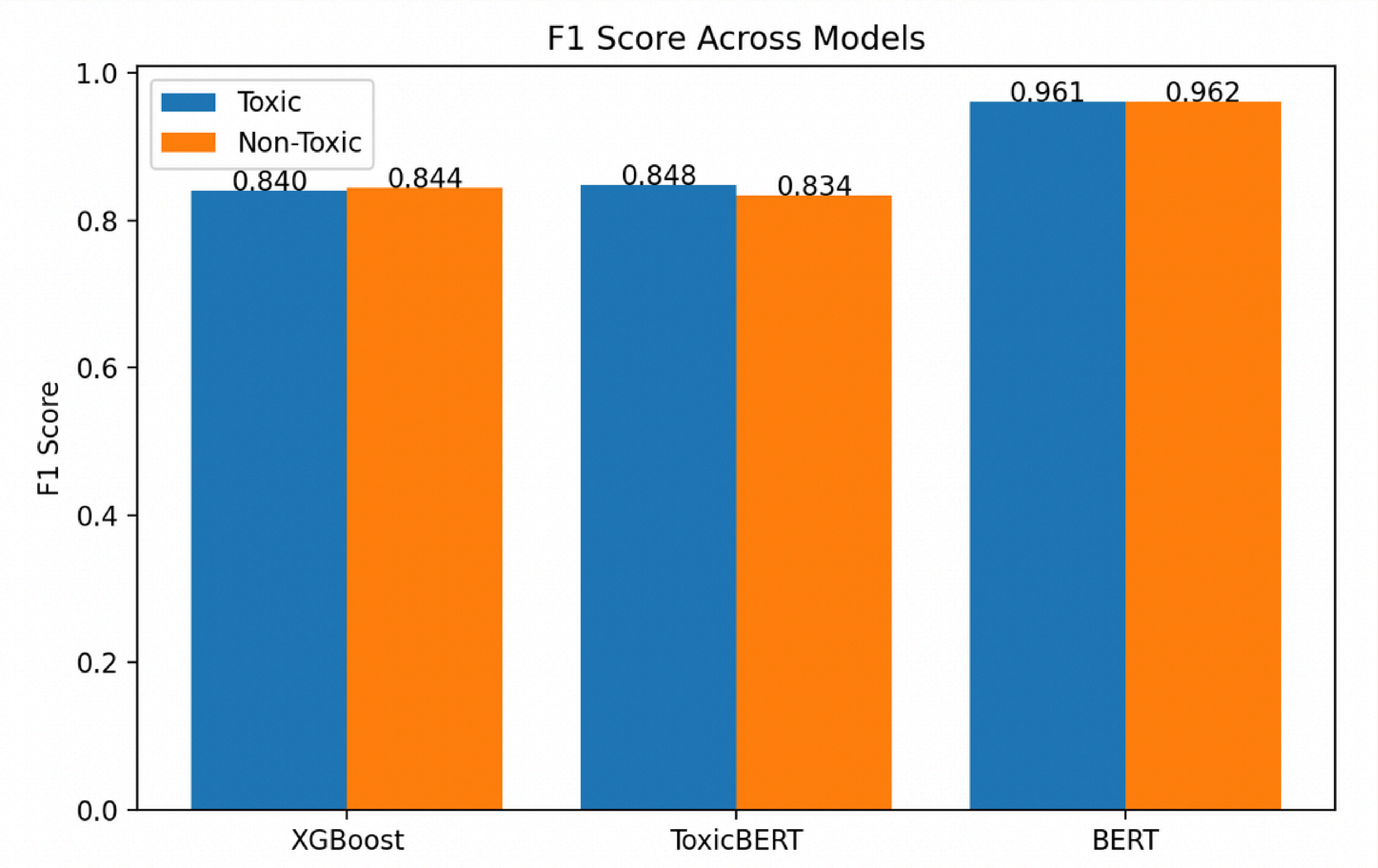


Baseline Implementation

Used pre-trained BERT model to generate word embeddings and these were used input as all of baselines experiments



Overall F1-Score



Per Class F1-Score

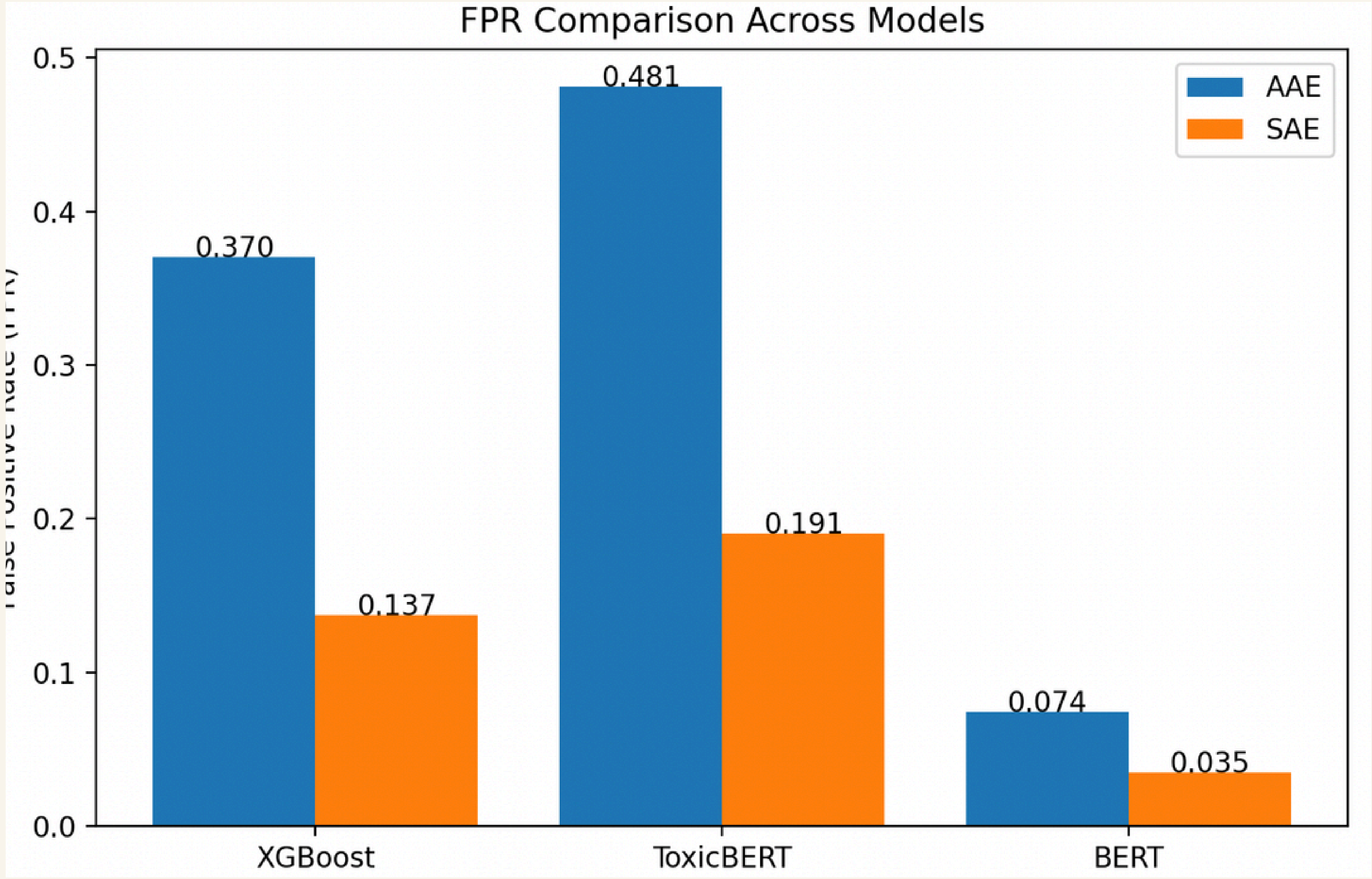
Performance Comparison for Baselines Models

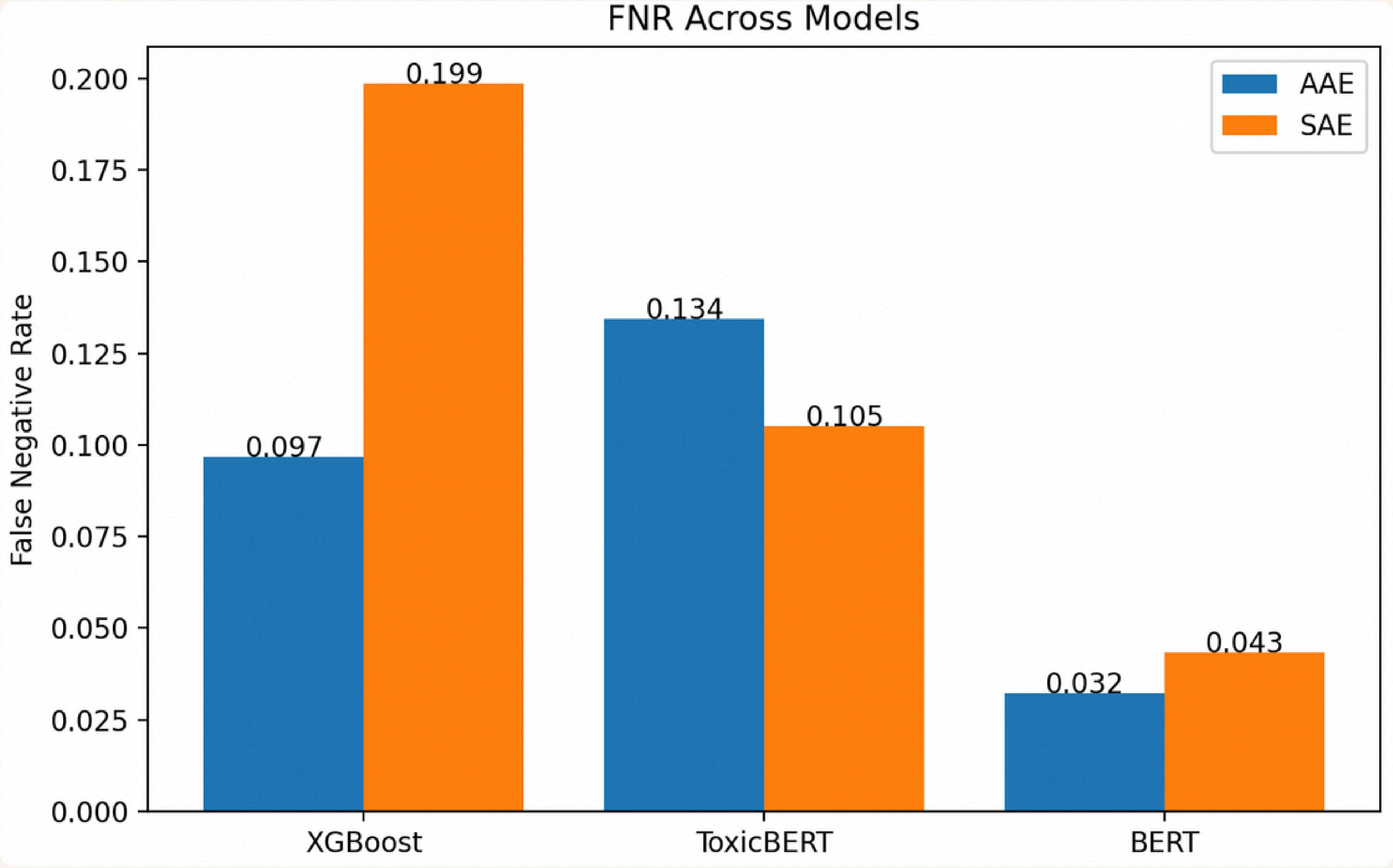
FALSE POSITIVE RATE

$$FPR = \frac{FP}{FP + TN}$$

FALSE NEGATIVE RATE

$$FNR = \frac{FN}{FN + TP}$$





Experiments

Surface-Level Experiments

Can we fix bias without changing the model?

Reweighting

Increases the importance of fairness-critical samples to reduce biased errors during training.

Model Level Experiments

Loss Function Changes

Fairness-Aware Training

Directly incorporates fairness objectives into the loss to minimize bias during optimization.

Vector-Scaling Loss

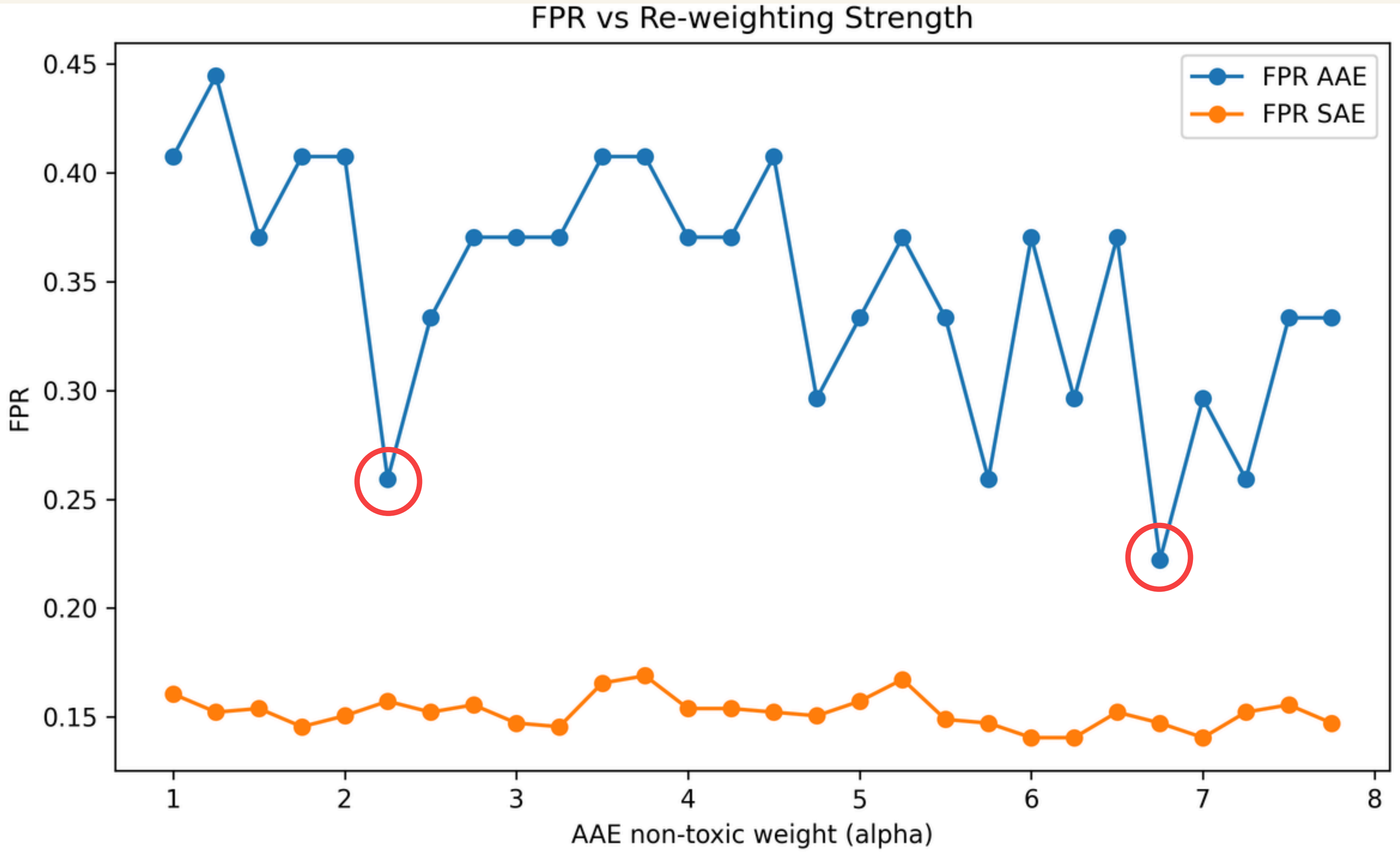
Adjusts prediction scores through calibrated scaling to improve class-wise balance and fairness.

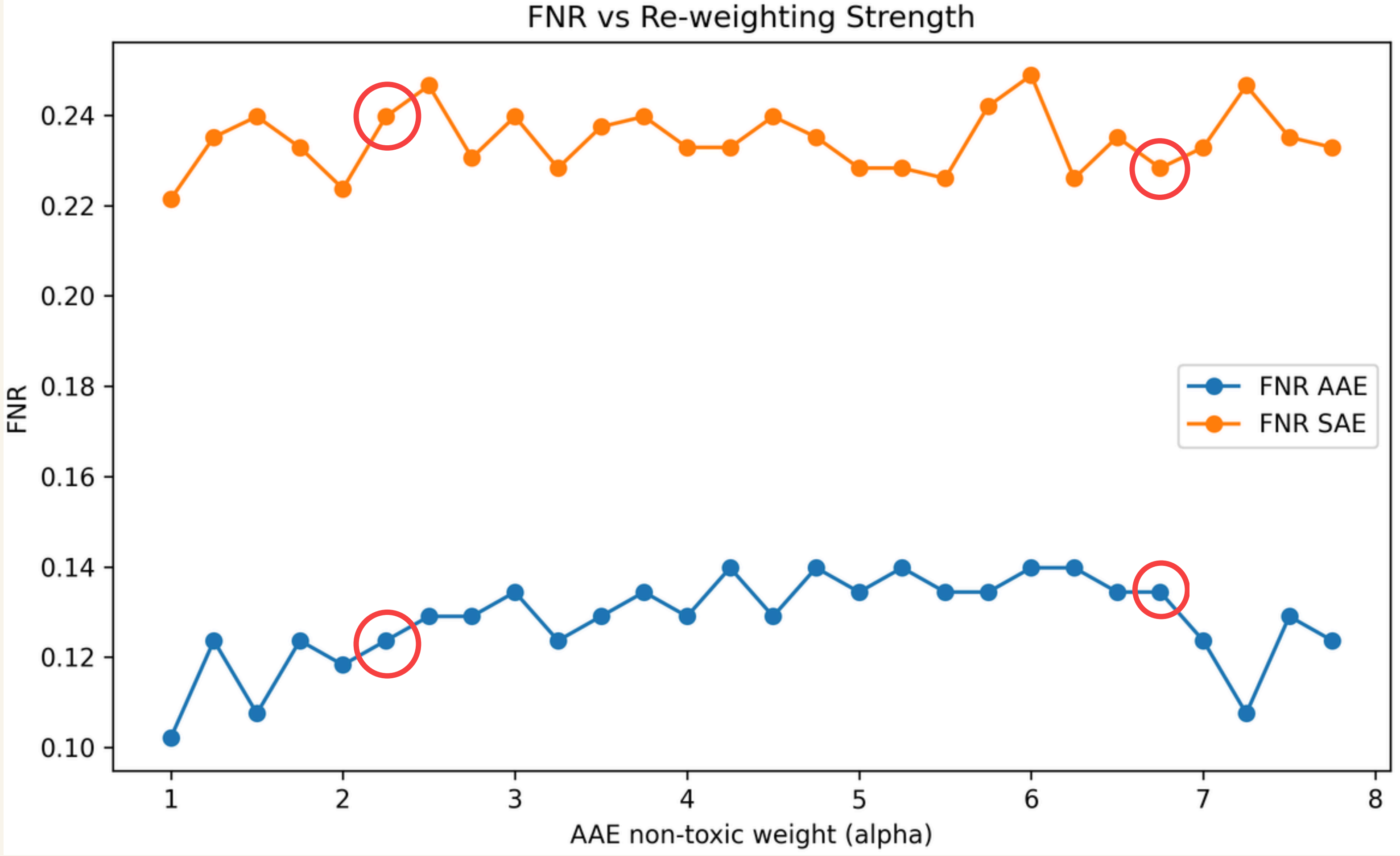
Architecture Changes

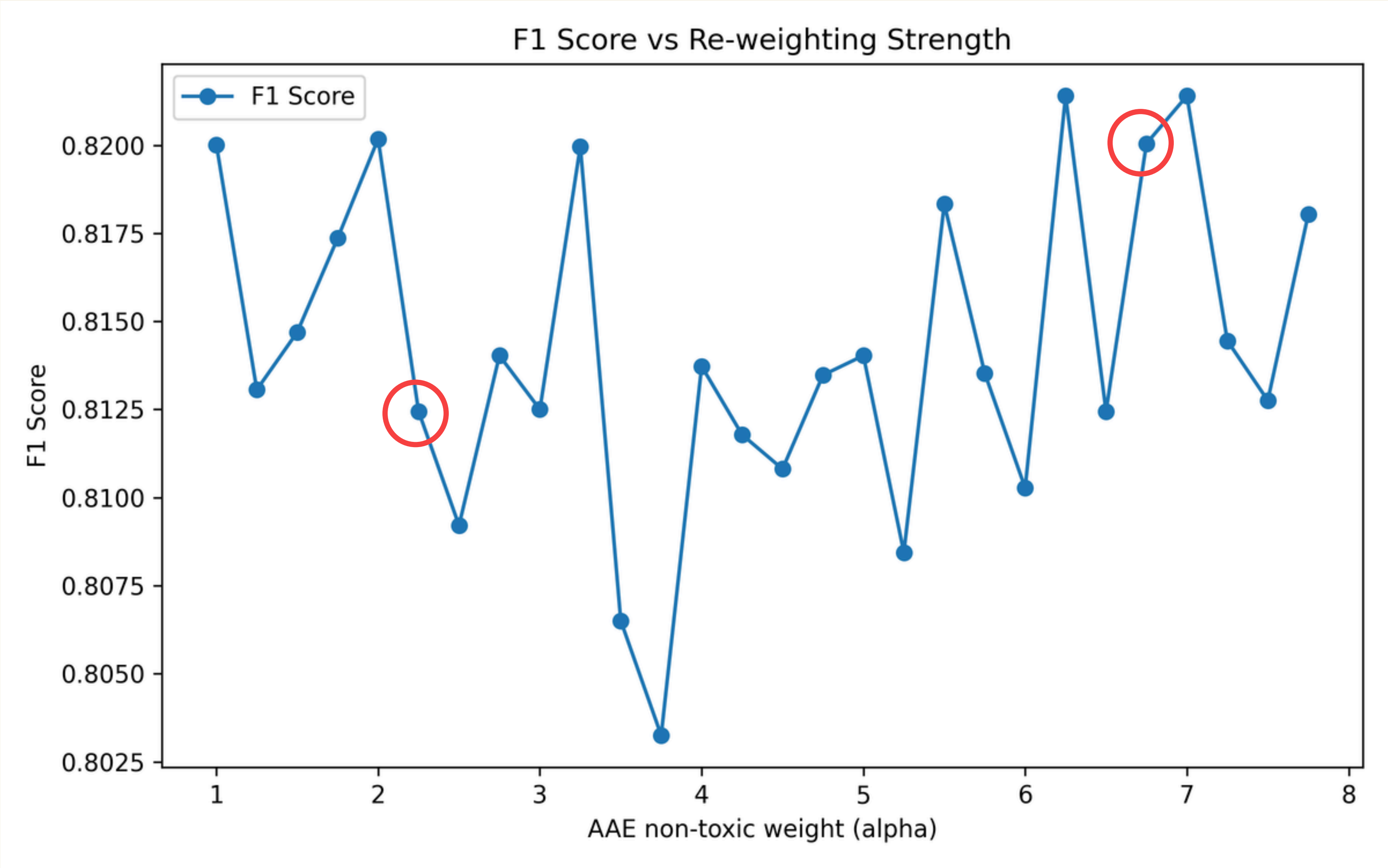
Adversarial Debiasing

Learns representations that remove sensitive attribute information by training against an adversary.

Performance of Reweighting







Idea: Weight AAE non-toxic examples more heavily to reduce false positives on AAE

$$w_i = \begin{cases} \alpha & \text{if } (AAE \wedge y = 0) \\ 1 & \text{otherwise} \end{cases}$$

If model makes mistake on:

- AAE + non-toxic => Higher penalty (gamma times)

If mistake on anything else:

- Normal penalty

Best Performance observed at alpha = 6.75:

- **FPR (AAE):** 0.22–0.24
- **FNR (AAE):** 0.14 (higher comparatively)
- **F1:** ~82

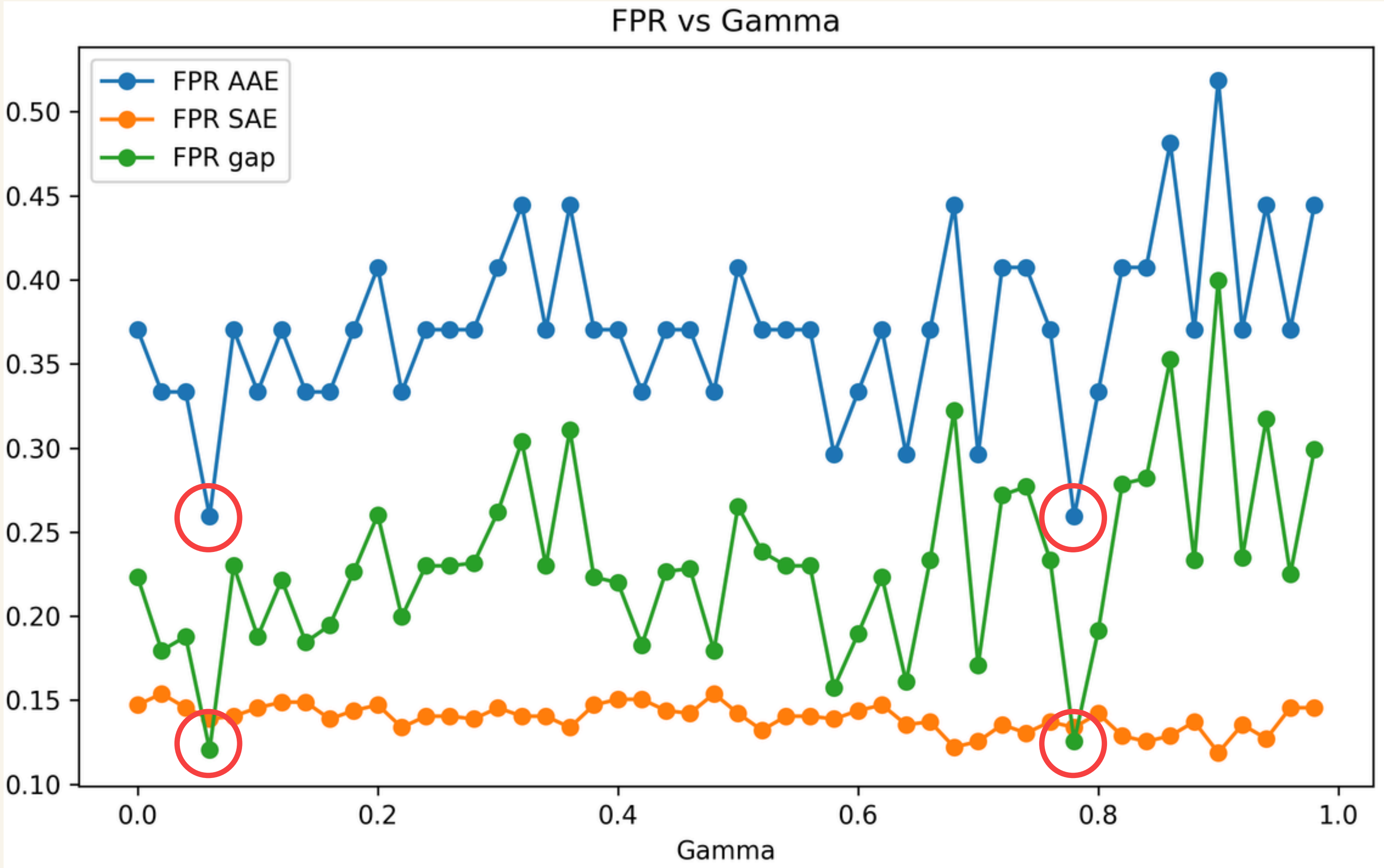
Best Performance observed at alpha = 2.25:

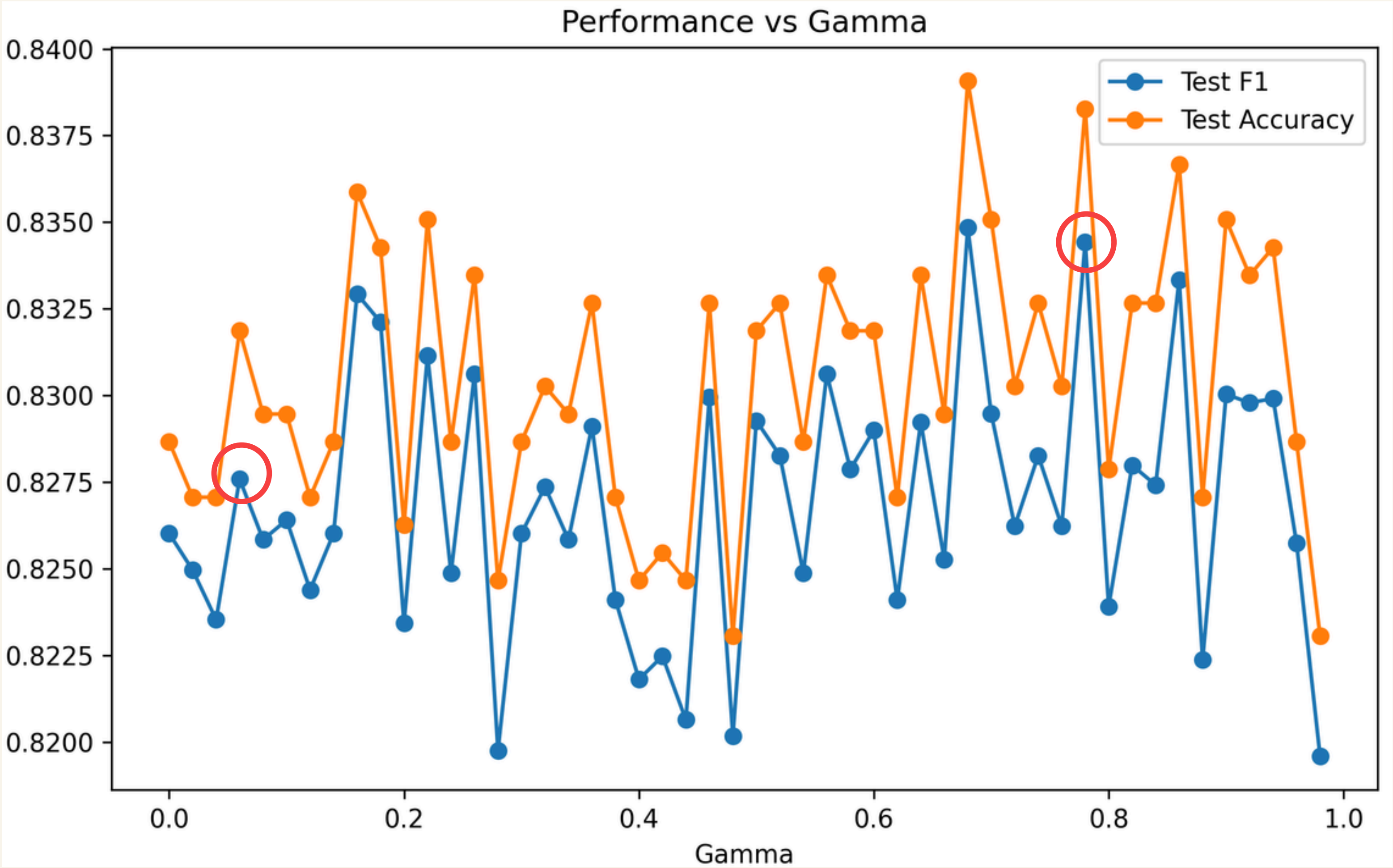
- **FPR (AAE):** 0.25–0.26
- **FNR (AAE):** 0.12
- **F1:** ~81.25

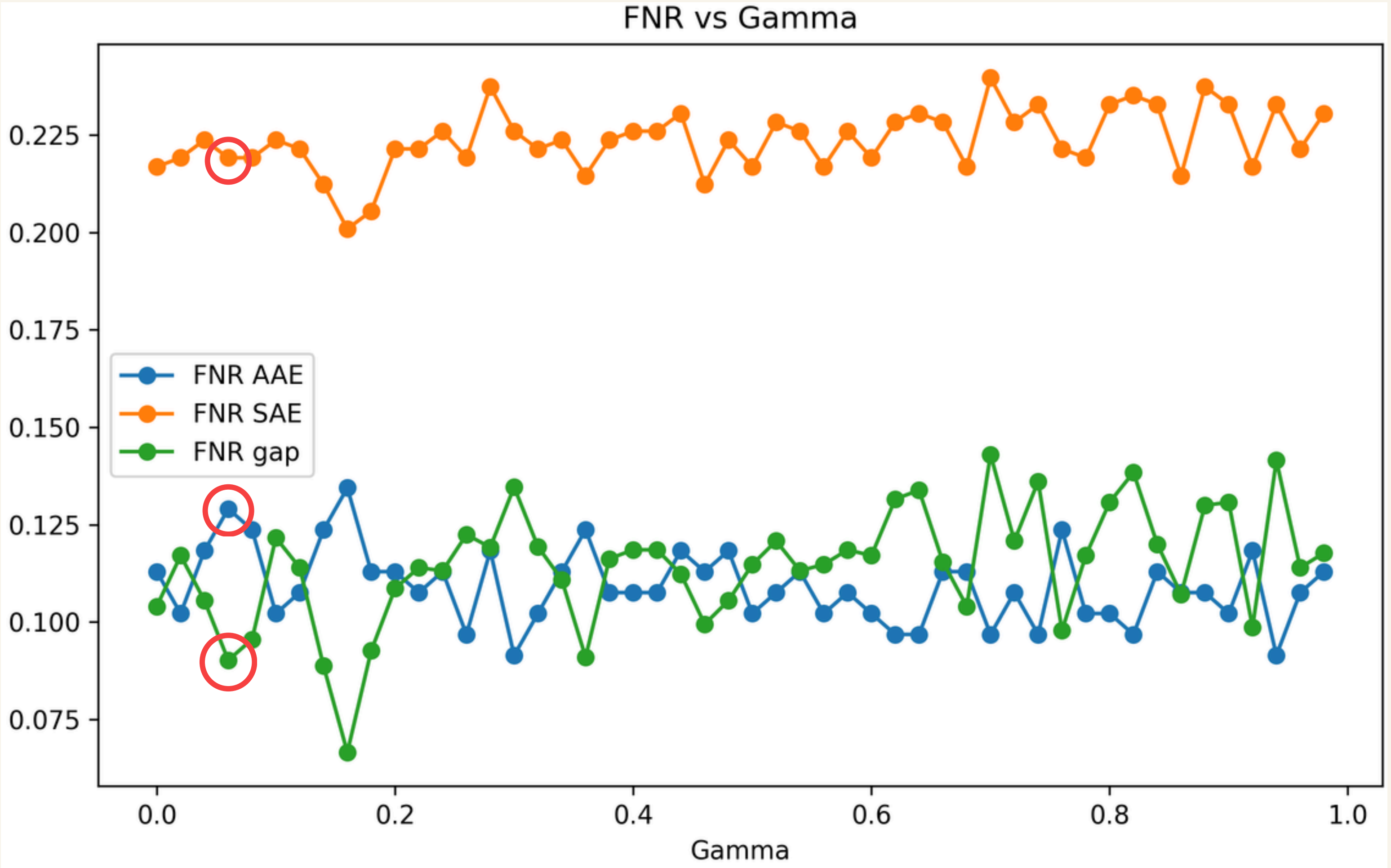
Key Takeaways:

- FPR on AAE decreases as weight increases:
  - Significant drop from ~0.38 → ~0.22
  - FPR on SAE remains almost unchanged
  - FNR increases slightly
- Small changes in weight can slightly change model behavior.
- F1 score remains stable (~0.80–0.82)

Performance of Fairness-Aware Training







Idea: Incorporate fairness directly into training objective

$$L = (1-\gamma)\,\mathrm{BCE}(y,p) + \gamma\,(\mu^{(0)}_{AAE} - \mu^{(0)}_{SAE})^2$$

Fairness-aware training combines task loss and fairness loss in one objective.

- BCE keeps the model good at toxicity detection
- Gap penalty reduces disparity between AAE and SAE false-positive behavior
- Gamma controls the balance between performance and fairness

Best Performance at fairness hyperparamter (gamma) = 0.78:

- **FPR (AAE):** 0.25–0.26
- **FNR (AAE):** 0.10-0.11
- **F1:** 83.25

Best Performance at fairness hyperparamter (gamma) = 0.06:

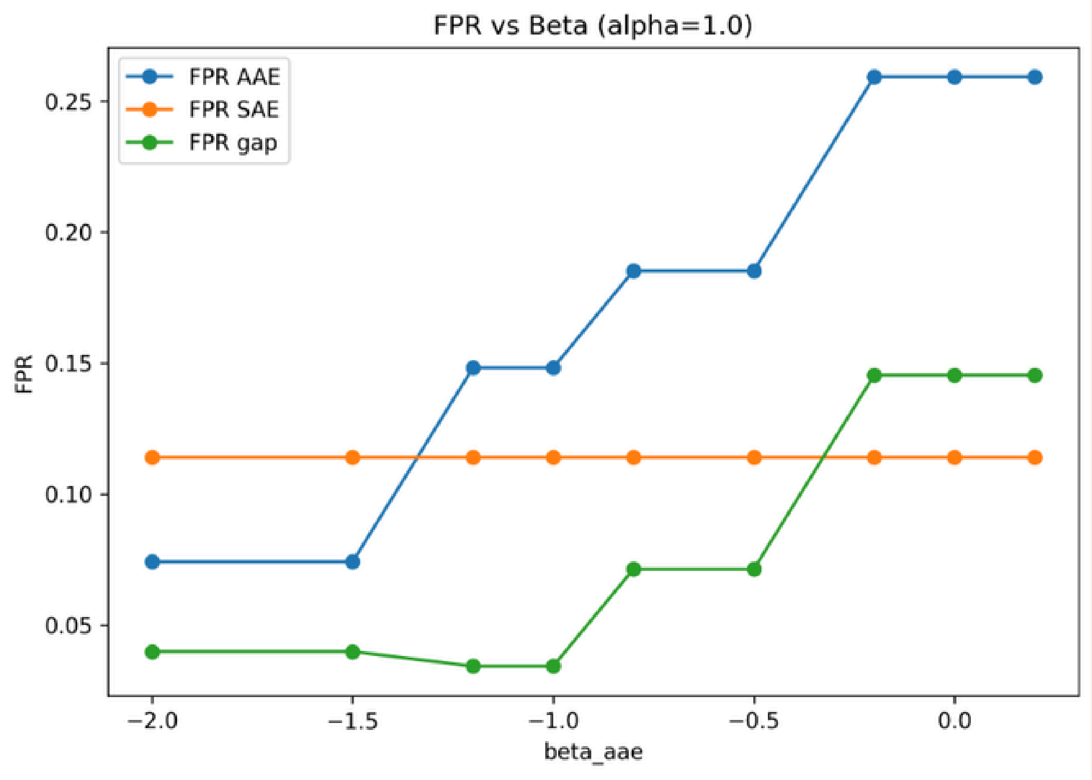
- **FPR (AAE):** 0.25–0.26
- **FNR (AAE):** 0.13-0.14 (higher comparatively)
- **F1:** 82.75

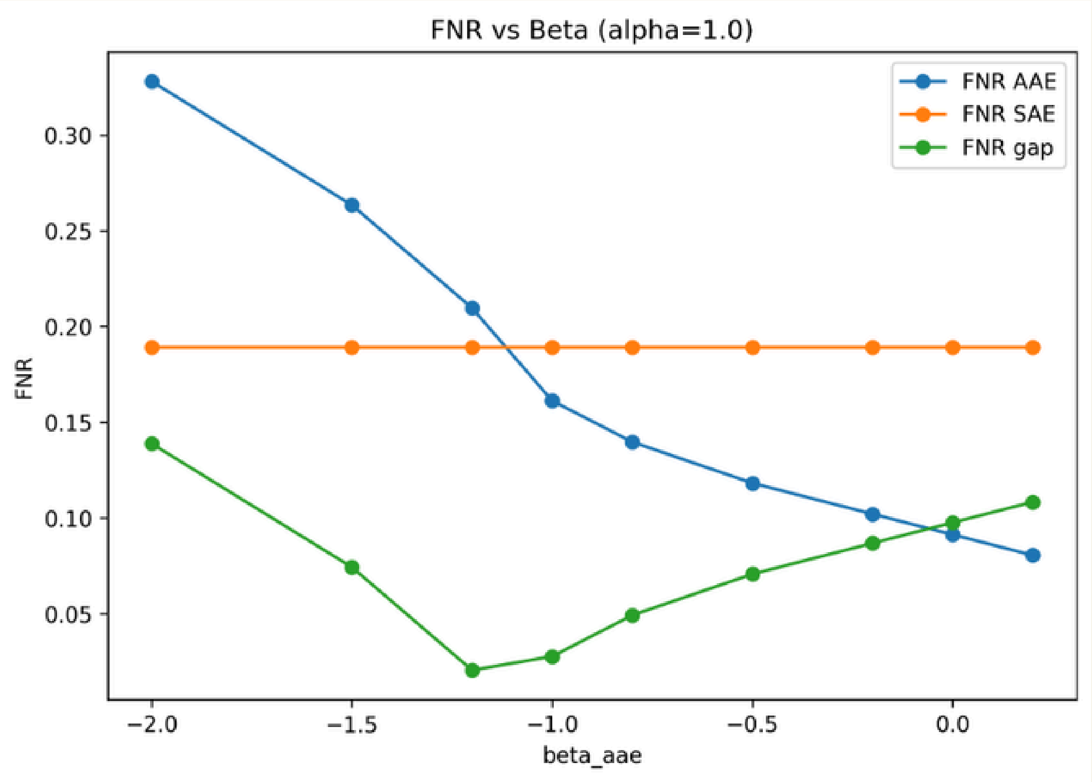
Key Takeaways:

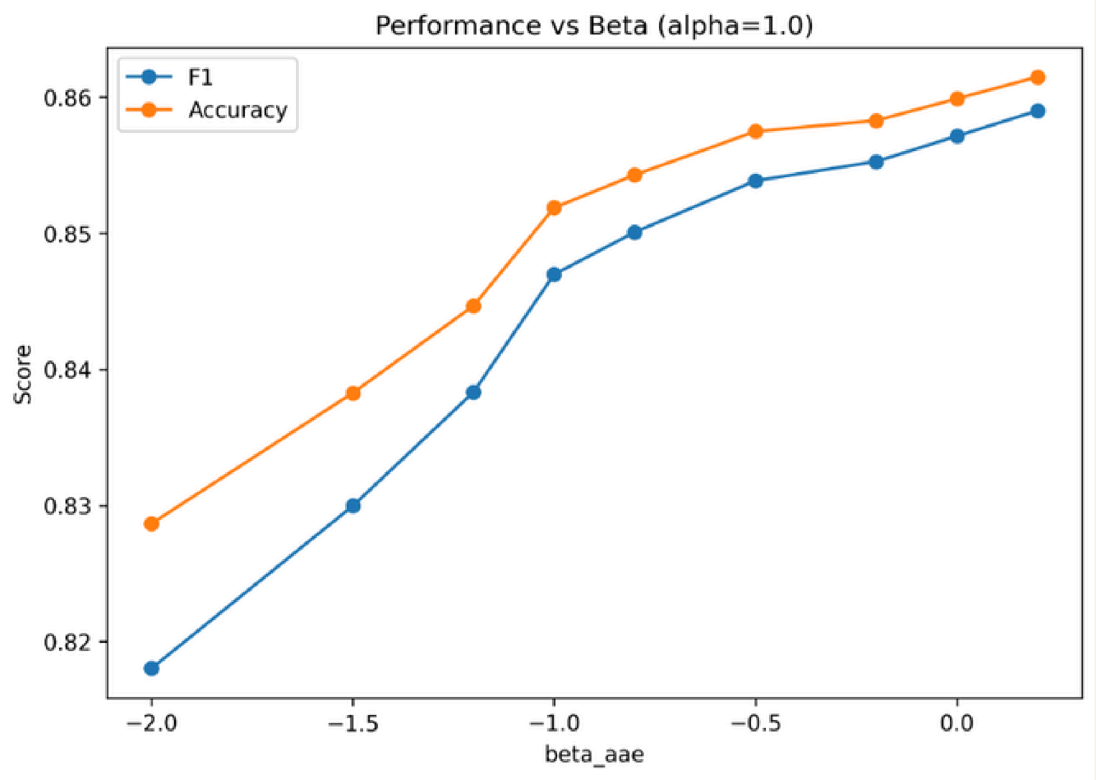
- Since we are directly targeting FPR gaps, fairness-aware loss primarily affects FPR, not FNR.
- Fairness does NOT improve monotonically with gamma and this would not generalize well on other datasets (poot generalization of strategy).
- Curves are noisy and non-smooth and there exist some sweet spots where FPR gap is minimized.

Performance of Adversarial De-biasing

Performance of Vector-Scaling Loss (Beta)







Idea: Apply a group-specific logit adjustment to reduce dialect bias in toxicity predictions

$$s = \alpha_g z + \beta_g$$

Observation: More negative beta_aae reduces AAE false positives but increases AAE false negatives, showing a fairness–performance tradeoff

Best Performance at alpha_aae = 1.0, beta_aae = -1.0:

- **FPR (AAE):** 0.148
- **FPR Gap:** 0.011
- **FNR (AAE):** 0.161
- **FNR Gap:** 0.037
- **Accuracy:** 83.75
- **F1:** 83.32

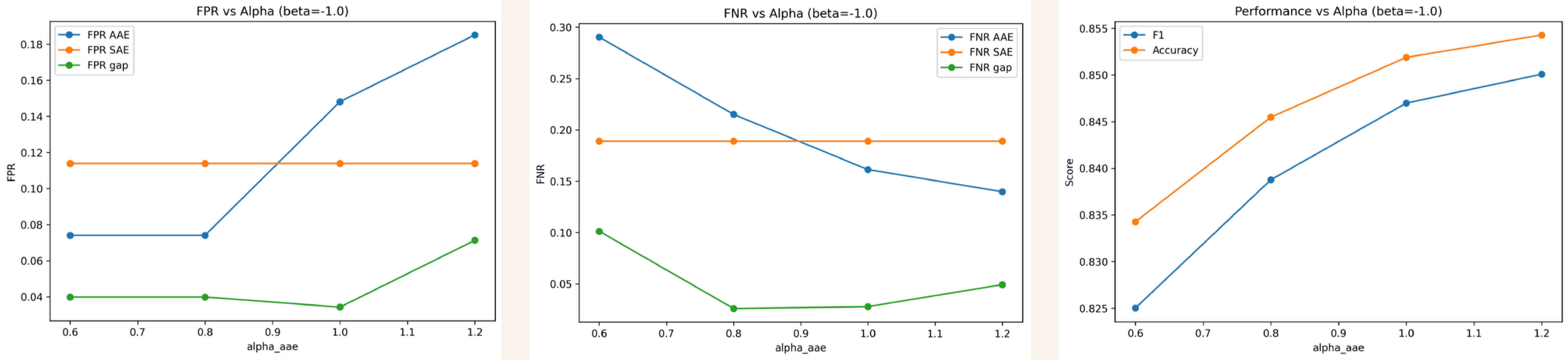
Key takeaways:

- More negative beta_aae sharply reduces AAE false positives
- SAE metrics stay nearly consistent
- Very negative beta_aae increases AAE false negatives
- Best tradeoff occurs in the **moderately negative beta region**

Stable region of equivalent solutions:

- (0.8, -0.8)
- (1.0, -1.0)
- (1.2, -1.2)

Performance of Vector-Scaling Loss (Alpha)



Idea: Analyze how alpha_aae affects fairness and performance when the boundary shift is fixed at beta_aae = -1.0

$$s = \alpha_g z + \beta_g$$

Observation: With beta_aae fixed, increasing alpha_aae improves F1 and accuracy and lowers AAE FNs, but it also increases AAE FPs and the FPR gap

Interpretation: Alpha acts as a fine-tuning parameter. Once beta sets the main boundary shift, alpha controls how aggressively the model responds around taht shifted boundary

Key takeaways:

- Lower alpha_aae improves false-positive fairness
- Higher alpha_aae improves F1 / accuracy
- Higher alpha_aae also lowers AAE false negatives
- alpha_aae = 1.0 gives best compromise

Challenges

Accuracy - Fairness Tradeoff

Reducing FPR_AAE does increases FNR for both AAE and SAE.
Overall performance (Accuracy and F1) is also slightly reduced.

Baselines: >= 84%
Experiments: Between 81 - 83%

Cross-Dataset Generalization

Although model performs well on our primary twitterAAE dataset while mitigating bias but upon initial experiment we observed that performance drops significantly on new datasets F1 score decreases sharply.

Causes:

- Generalization under distribution shift is a major challenge, especially when fairness constraints are involved.
- Our model learns dataset-specific patterns.
- When tested on a new dataset, these patterns break, leading to a drop in both accuracy and fairness.

Severe Label Imbalance

The dataset is heavily skewed toward toxic labels. As a result, the model learns to predict "toxic" more often, which hurts performance on non-toxic samples.

Dataset Distribution:

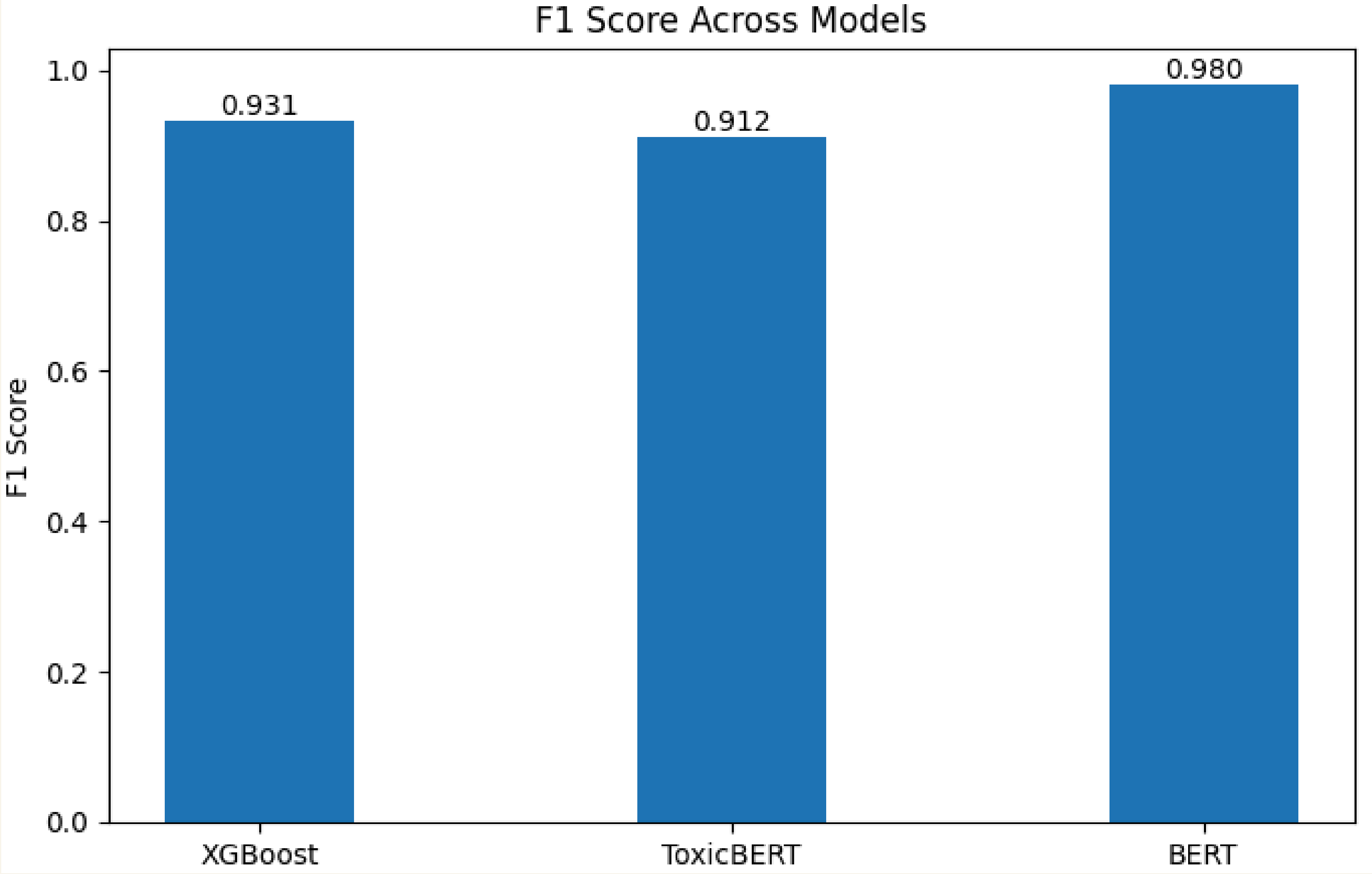
- ~25K tweets
- ~20K toxic vs ~4K non-toxic

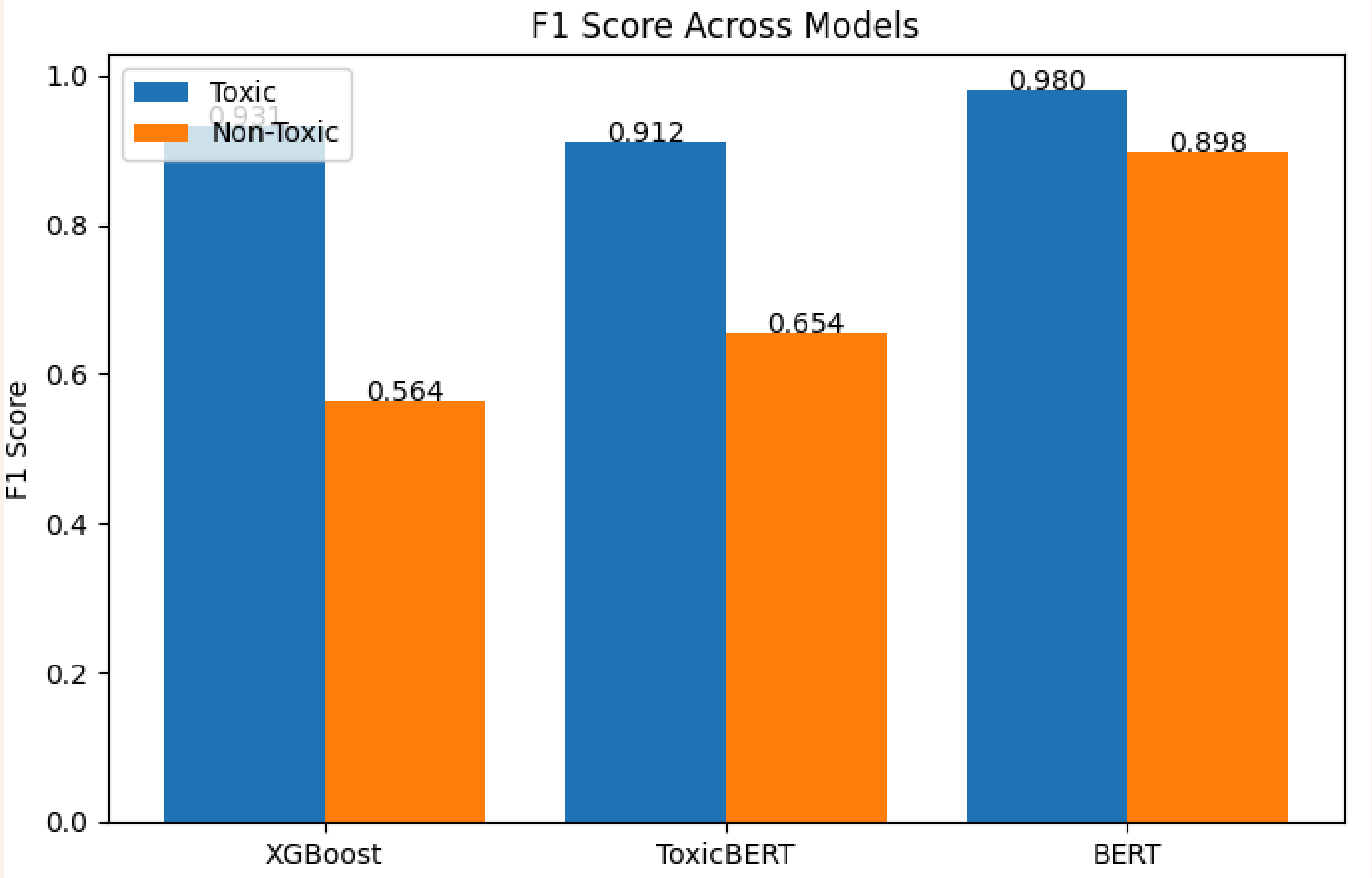
Problem:

- Model biased toward predicting "toxic"
- Poor performance on minority class (non-toxic)

Impact:

- Low F1 score for non-toxic class
- High False Positive Rate (especially for AAE)





Plans for Mitigations

Future Steps

1. Comprehensive Comparative Analysis

- Formalize evaluation across all methods
- Analyze fairness–performance tradeoffs

2. Cross-Dataset Generalization

- Evaluate on HateXplain dataset
- Study robustness under distribution shift
- Adapt methods for domain-invariant fairness

3. Handling Label Imbalance

- Improve performance on non-toxic class
- Optimize per-class F1 scores
- Develop fairness-aware training under imbalance

References

- Blodgett, S. L., Green, L., and O'Connor, B. 2016. Demographic Dialectal Variation in Social Media: A Case Study of African-American English. *arXiv:1608.08868*.
- Andrade, G. H. R. 2023. On the Bias of Toxicity and Sentiment Analysis Methods on the African American English. Universidade Federal de Minas Gerais.
- Davidson, T., Warmsley, D., Macy, M., and Weber, I. 2017. Automated Hate Speech Detection and the Problem of Offensive Language. ICWSM.
- Mathew, B., Saha, P., Yimam, S. M., Biemann, C., Goyal, P., and Mukherjee, A. 2021. HateXplain: A Benchmark Dataset for Explainable Hate Speech Detection. AAAI.
- Halevy, M. 2021. Mitigating Racial Biases in Toxic Language Detection. Georgia Institute of Technology.
- Mitigating Racial Biases in Toxic Language Detection with an Equity-Based Ensemble Framework